

Assignment 10

1. Name the various types of diverging sections possible for a c-d nozzle.
2. A conical nozzle of area ratio 20 has half-angle of 15° . What is the length of the nozzle in terms of its throat diameter? What is its divergence loss?
3. Describe a maximum-thrust nozzle. What is its alternate name? Does the flow leave this type of nozzle axially?
4. What is a perfect nozzle? Name its disadvantage.
5. Draw the shock-expansion pattern in a moderately over-expanded exhaust jet plume. How will the flow behavior change were the jet to become severely over-expanded?
6. Draw the shock-expansion pattern in an under-expanded exhaust jet plume.
7. Briefly describe the plug nozzle. What disadvantage of a bell nozzle does it overcome? What is its disadvantage?